

Kiyan Amirian

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Research Summary

PhD student in Materials Science and Engineering at the University of Maryland, working at the intersection of machine learning and computational materials science. Research focuses on diagnosing where machine-learned interatomic potentials fail in realistic atomistic simulations, spanning force prediction, phase stability, elemental ordering, and ion migration, to guide the development of more reliable models.

Education

University of Maryland

Doctor of Philosophy in Materials Science and Engineering

Cumulative GPA: 4.0/4.0

College Park, MD, USA

2023–Expected Aug 2027

Shiraz University

Bachelor of Science in Materials Science and Engineering

Cumulative GPA: 17.15/20 (3.5/4.0); Last 2-year GPA: 18.54/20 (3.9/4.0)

Shiraz, Fars, Iran

2018–2023

Research Interests

Computational Materials Science

Atomistic Simulation

Machine-Learned Interatomic Potentials

AI for Science

Publications

K. Amirian, R. S. Saravanan, F. Adams, C. E. Schwarz, Y. Mo

arXiv, 2026

FP-DeErr: Application-Oriented Error Decomposition for Foundation Potentials

[arXiv:2609.05714](#) · [GitHub](#) · [Leaderboard](#)

K. Amirian, Z. Abbasi, R. Ebrahimi

2023

Analytical and numerical instability analysis of functionally graded low-carbon steel.

Steel Symposium 401, International Journal of Iron and Steel Society of Iran.

[doi:10.22034/IJISSI.2023.1990580.1262](https://doi.org/10.22034/IJISSI.2023.1990580.1262)

Research Experience

Graduate Researcher, UMD (Advisor: Prof. Yifei Mo)

2024–Present

- Developed FP-DeErr, an open-source, application-oriented error-decomposition framework for foundation potentials (FPs) that resolves errors by the physical quantities, configurations, and workflow stages governing each computational task; built its [evaluation code](#) and [public leaderboard](#), benchmarking 18+ FPs against DFT with ASE, MatCalc, and pymatgen.
- **Force prediction:** decomposed force errors into magnitude and angle, and designed metrics for highly accurate predictions, large-force-error atoms, and far-from-equilibrium atoms (MatPES, OMat24), showing that these failure modes are diluted in average MAE/RMSE.
- **Phase stability and elemental ordering:** designed convex-hull agreement metrics (ground state, within-phase and global hull minimum) and evaluated ordering rankings (Top-1, Recall@k, Spearman's ρ) on chalcogenide phase-change materials, separating intrinsic energy-ranking errors from FP-relaxation errors.
- **Ion migration:** benchmarked FP-NEB against DFT-NEB on 154 Li- and Na-ion pathways with metrics for convergence, profile shape, endpoint energy ranking, and barriers, tracing barrier errors to intrinsic PES error, endpoint relaxation, and along-path forces.

Undergraduate Researcher, Shiraz University (Advisor: Prof. Ramin Ebrahimi)

2019–2022

- Developed an algorithm to predict tensile instability in functionally graded low-carbon steel and compared its predictions with published experimental data ([paper](#)).

Selected Projects

KV Cache Optimization for Large Language Models

Fall 2025

Led a team comparing KV-cache compression techniques (StreamingLLM, H2O, KIVI, Zip-Cache) for long-context LLM inference. Reimplemented KIVI 2-bit quantization for LLaMA-2 and designed evaluation workflows across summarization, reasoning, and QA benchmarks, adding metrics for token match rate, entity hallucination, and sentence compliance ([GitHub](#)).

Image Classification: Classical ML and Transfer Learning

Spring 2025

Built pipelines for MNIST and a 10-class species dataset using logistic regression from scratch, RBF SVM+PCA on MNIST (97.1%), and frozen-backbone VGG19 transfer learning ([GitHub](#)).

Face Recognition: Dimensionality Reduction and Kernel Classifiers

Spring 2025

Benchmarked PCA/MDA with Bayes, k-NN, SVM, and AdaBoost on a 200-subject dataset. Implemented a custom gradient-descent kernel SVM to address CVXOPT numerical failures with high-degree polynomial kernels ([GitHub](#)).

Powder Bed Fusion Thermal Simulation

Spring 2022

Implemented custom FORTRAN subroutines in ABAQUS to model laser-material interaction including spatially varying heat flux and temperature-dependent material properties.

Skills

Programming: Python (NumPy, Matplotlib), Bash, MATLAB, Fortran, Git, $\LaTeX$

Atomistic Simulation: DFT (VASP), LAMMPS, ASE, pymatgen, MatCalc; NEB, molecular dynamics

Foundation Potentials (MLIPs): MACE, CHGNet, M3GNet, UMA, Orb, MatterSim, DPA4, eSEN

ML Frameworks: PyTorch, TensorFlow/Keras, scikit-learn

ML Methods: Transfer learning, CNNs, SVMs, LLM KV-cache quantization

AI Tools: LLM coding agents (Claude, ChatGPT) for developing, debugging, and validating research code

HPC: SLURM job arrays on university HPC clusters

Continuum Modeling: Abaqus (FEA)

Honors & Awards

Department of Materials Science and Engineering, University of Maryland

College Park, MD

University of Maryland Dean's Fellowship Award, \$43,000

Spring 2023

Department of Materials Science and Technology, Shiraz University

Shiraz, Fars, Iran

Ranked 1st among 56 students in B.Sc. Materials Science and Engineering

Spring 2022

Relevant Graduate Courses

Introduction to Atomistic Modeling (4.0/4.0)

Natural Language Processing (4.0/4.0)

Statistical Pattern Recognition (4.0/4.0)

Machine Learning for Materials Science (4.0/4.0)

Teaching Experience

Kinetics, Diffusion and Phase Transformations (Fall 2024)

Mechanical Properties II (Spring 2022)

Fundamentals of Computer (Spring 2020)

Algebra (Fall 2019)

References

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Haizhao Yang

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